Qwen

An Executable Constitution for the Home: A Firmware-Based Safety Kernel for Integrating Legacy Appliances into Smart-City Infrastructures

A Formalized Intent-Based Safety Kernel for Heterogeneous Hardware

The development of a universal firmware-based framework hinges on a foundational principle of abstraction: separating high-level intent from low-level actuation. This approach allows a single, standardized control logic to manage a heterogeneous collection of legacy household appliances, treating them as interchangeable nodes within a larger smart-city infrastructure . The core of this strategy is a formalized safety kernel implemented as a tiny, universal Application Programming Interface (API) that operates at the firmware level . This API defines a minimal set of parameters for a pest-deterrent action, thereby insulating the decision-making process from the physical idiosyncrasies of any given device. The proposed `PdssIntent` struct serves as the primary interface between the central controller and the appliance adapter, encapsulating all necessary information for an actuation event in a vendor-agnostic manner . Its fields—`profile`, `intensity01`, `duty01`, and `period_seconds`—represent a high-level description of the desired effect rather than raw electrical signals . For instance, instead of sending a command to "turn on relay X for Y milliseconds," the system sends an intent to execute the profile `"bedbug_cool_surface_low"` at 75% intensity with a 30% duty cycle . This abstraction is the critical enabler of universal compatibility, allowing the same control logic to be applied to a 1980s box fan and a 2024 smart air conditioner alike .

This intent-based model is underpinned by a mathematically rigorous safety invariant designed to prevent unsafe system states. The `safestep` function, a core component of the firmware kernel, acts as the gatekeeper for all actuations, enforcing two hard constraints before any physical change is permitted . The first constraint is the enforcement of normalized risk corridors. Each potential emission from an appliance—acoustic noise, thermal radiation, and structural vibration—is mapped onto a normalized coordinate (`rnoise`, `rthermal`, `rvib`). These coordinates are calculated by comparing

a measured telemetry value (e.g., decibels, surface temperature rise, vibration root mean square) against predefined safe, gold, and hard corridor bands . If any resulting `rj` coordinate reaches or exceeds 1.0, indicating a breach of the hard limit, the `safestep` function immediately returns a failure status, blocking the intended action . This ensures that no single actuation can push the appliance beyond its pre-defined human and structural safety limits.

The second, more dynamic constraint is the Lyapunov residual stability condition. The kernel computes a scalar residual value, V_t, which is a weighted sum of the squares of the normalized risk coordinates: $V_t = \sum w_j r_j^2$. This residual represents the system's distance from a perfectly safe state (where all `rj` are zero). The `safestep` function verifies that the residual for the next time step, V_{t+1}, is not greater than the current residual, V_t . By enforcing the invariant $V_{t+1} \leq V_t$, the system guarantees that it cannot drift away from the viability kernel—the region of operation where all risks are deemed acceptable. This provides a closed-loop stability guarantee, making unsafe trajectories mathematically impossible . The combination of hard corridor limits and a non-increasing Lyapunov residual creates an "executable constitution" for the system, a set of unchangeable rules that apply equally to every connected device, regardless of its age or internal complexity . This formal safety model is the cornerstone upon which the entire framework of repurposing legacy hardware is built, ensuring that the pursuit of adaptive pest deterrence does not compromise occupant safety or property integrity.

To operationalize this safety kernel, a formal data structure is required to represent the state of each appliance node. The proposed `ApplianceNodeShard2026v1` schema, defined in ALN syntax, provides this persistent, structured representation . It contains fields for unique identification (`nodeid`, `houselabel`, `roomlabel`), appliance metadata (`applianceclass`, `ratedpowerW`), and crucially, the governance and telemetry data that feed the safety kernel . The shard explicitly stores the corridor bands for each risk channel (`noise_safe_dBA`, `dT_gold_C`, etc.), the currently computed normalized risk coordinates (`rnoise`, `rthermal`, `rvib`), the Lyapunov residual (V_t), and the governance scores (Knowledge-factor, Eco-impact, Risk-of-harm) . Furthermore, it includes fields for tracking usage statistics, such as `pdss_duty_fraction` and `pdss_energy_kWh_per_day`, and anchors for evidence, linking the node's configuration and residuals to verifiable transactions on a distributed ledger like Bostrom . This shard serves multiple purposes: it is the runtime state of the node, a log for auditing decisions, and the basis for city-scale aggregation and impact analysis. By defining this canonical shard format, the framework establishes a common language for all nodes, enabling consistent monitoring and governance across the entire network of legacy appliances . The use of Rust and ALN for implementation aligns with a focus on

both robust, memory-safe execution in embedded environments and verifiable, formal governance [15].

The following table details the proposed `ApplianceNodeShard2026v1` schema, outlining its fields, data types, and descriptions based on the provided context.

Field Name	Data Type	Description
nodeid	string	Unique identifier for the appliance node.
region	string	Geographic stamp, e.g., US-AZ-Phoenix.
houselabel	string	Appliance/house identifier.
roomlabel	string	Specific room, e.g., Bedroom-1, Kitchen.
applianceclass	string	Class of appliance, e.g., Fan, Fridge, WindowAC, TV, Audio.
adaptertype	string	Type of firmware adapter, e.g., SmartPlug, IRShim, PanelRelay.
ratedpowerW	float	Nominal power rating in Watts.
ratednoise_dBA	float	Nominal noise rating in decibels A-weighted.
noise_safe_dBA	float	Safe band upper limit for acoustic noise.
noise_gold_dBA	float	Gold band upper limit for acoustic noise.
noise_hard_dBA	float	Hard band upper limit for acoustic noise.
dT_safe_C	float	Safe band upper limit for surface temperature rise.
dT_gold_C	float	Gold band upper limit for surface temperature rise.
dT_hard_C	float	Hard band upper limit for surface temperature rise.
vib_safe	float	Safe band upper limit for normalized vibration.
vib_gold	float	Gold band upper limit for normalized vibration.
vib_hard	float	Hard band upper limit for normalized vibration.
rnoise	float	Current normalized risk coordinate for noise.
rthermal	float	Current normalized risk coordinate for thermal load.
rvib	float	Current normalized risk coordinate for vibration.
residualVt	float	Current Lyapunov residual, V_t.
knowledgefactor01	float	Normalized knowledge factor score in [0,1].
ecoimpact01	float	Normalized eco-impact score in [0,1].
riskofharm01	float	Normalized risk-of-harm score in [0,1].
pdss_duty_fraction	float	Fraction of time the appliance is under PDSS control.
pdss_energy_kWh_per_day	float	Energy consumed daily for pest deterrence.
baseline_pesticide_g_per_day_avoided	float	Estimated pesticide mass avoided per day.
bostrom_tx_hex	string	Hexadecimal anchor to a Bostrom transaction.
evidencehex	string	Hexadecimal evidence anchor for governance.
notes	string	Free-text field for additional notes.

This structured approach ensures that every node, whether it's a simple fan controlled by a smart plug or a complex window AC unit managed via an IR shim, is represented and governed by the same formal model. This uniformity is essential for scaling the system to a city-wide level while maintaining a consistent and verifiable standard of safety. The kernel's reliance on normalized coordinates and a scalar residual allows it to reason about risk in an abstract space, independent of the specific physics of how a particular appliance generates noise or heat . This abstraction layer is what makes the framework truly universal and resilient to the diversity inherent in legacy hardware.

Architectural Blueprint for Universal Connectivity and Adaptation

The architectural blueprint for integrating legacy appliances into the BugsLife PDSS framework is predicated on a flexible, multi-layered adapter system that bridges the gap between the abstract intent-based API and the physical world. The goal is to achieve universal compatibility through a small number of well-defined "adapter patterns" that can be deployed using either non-invasive or minimally invasive methods . This modular approach allows the core safety kernel to remain agnostic to the specific communication protocols and physical interfaces of the target appliances, focusing solely on enforcing the safety invariant . The proposed connectivity strategies span a spectrum from simple wall-plug interventions to more deeply integrated in-line modules, providing a comprehensive toolkit for retrofitting virtually any device found in a household.

Non-invasive methods form the first and most accessible layer of this architecture. These techniques leverage existing interfaces and power connections to exert control without altering the appliance's internal components. The most common example is the **smart plug**, which acts as a remotely controllable relay. When placed between an appliance and the mains outlet, a smart plug firmware agent can receive a `PdssIntent` from the edge gateway, compute the corresponding `safestep` decision, and then simply toggle its internal relay to turn the appliance on or off according to the specified duty cycle . This method is universally applicable but offers limited granularity, typically restricted to full-on or full-off states. Another key non-invasive technique is the use of **IR blasters**. Many older consumer electronics, such as televisions, audio systems, and even some window AC units, rely on infrared remote controls for operation [18] [125]. An IR blaster, acting as a firmware adapter, can learn the command sequences for various functions (e.g., power on, fan speed change, mode selection) and re-transmit them upon receiving a corresponding intent [18]. This allows for the emulation of user interactions, enabling fine-

grained control over devices that would otherwise be inaccessible. Finally, control can be established at the **breaker panel level**, where agents monitor and control power to entire circuits or specific rooms, offering a higher-level, building-wide coordination capability .

For applications requiring more direct or reliable control, the framework incorporates minimally invasive techniques. These methods involve creating a physical connection to the appliance's circuitry or internal bus, offering greater precision and potentially access to diagnostic data. **In-line relays** provide a robust, isolated means of switching an appliance on and off, often incorporating optocouplers to prevent electrical noise from propagating back into the control system [98]. This enhances the stability and reliability of the on/off control offered by smart plugs. A more advanced minimally invasive approach involves creating **virtual serial buses**. Many modern appliances, including refrigerators and HVAC units, contain internal microcontrollers communicating over low-speed buses like RS-485 or I²C [10] [11]. A firmware adapter can be designed to tap into these buses, effectively becoming a "shim" that intercepts and translates commands. For example, an adapter module could emulate a sensor on an I²C bus to trigger a specific response from the host appliance's controller, or it could inject Modbus RTU frames onto an RS-485 line to directly control a compressor or fan motor [98] [192]. This approach provides the most direct form of control, bypassing unreliable IR signals and coarse power cycling. The framework's design anticipates the discovery of new methods for virtual serial-bus creation, ensuring adaptability as new legacy devices are encountered .

A key innovation that elevates this framework beyond a static library of adapters is the concept of **firmware-adaptive connector profiles**. This mechanism addresses the challenge of recruiting very old or poorly documented devices whose exact emission characteristics are unknown . Instead of relying on manufacturer specifications or manual calibration for every single unit, the system employs an intelligent learning process. An edge gateway, aggregating control for a cluster of appliances in a home, can initiate a characterization phase for a new device . During this phase, the gateway systematically applies a range of gentle, low-duty-cycle `PdssIntents` and uses external sensors (e.g., a microphone, thermal camera, accelerometer) to collect telemetry data . For example, it might observe how a specific 1970s fan's acoustic footprint changes with different duty cycles or how a vintage refrigerator's compressor cycling affects the surface temperature of its chassis [177] [241]. This empirical data is then used to fit a conservative mathematical model, or "envelope," that maps duty cycle inputs to worst-case risk coordinate outputs (`rnoise`, `rthermal`, `rvib`) under local conditions .

This adaptive profile-building algorithm is designed to maintain a strict safety margin. Initially, the fitted envelopes will be conservative, assuming the worst-case behavior

within the observed data range . The `safestep` kernel will then operate using these learned, rather than theoretical, corridor bounds. As the system continues to operate and collects more telemetry over time, these envelopes can be progressively tightened, increasing the efficiency of the deterrent while remaining within the verified safe operating region . This iterative refinement process allows the framework to gradually onboard a vast inventory of legacy hardware, starting with safe, low-power patterns and building confidence as it learns each device's unique behavior. This learning capability is enabled by algorithms written in the same resource-constrained languages like Rust, avoiding heavyweight machine learning models in favor of lightweight statistical fitting and envelope-prediction techniques suitable for firmware . The resulting adaptive profiles are stored within the appliance's shard, making the knowledge portable and available to any gateway managing the device . This self-reliant, learning-based approach is the definitive solution for achieving true universal coverage, transforming the challenge of device heterogeneity from a barrier into an opportunity for data-driven optimization.

The implementation of this architecture relies on a combination of firmware libraries and specialized hardware modules. The core `LegacyAppliancePdssKernel-C++` library, containing the `PdssIntent` and `safestep` logic, can be cross-compiled for various microcontrollers used in smart plugs and low-end IoT devices [42] [45]. Bindings for common SDKs (e.g., ESP-IDF, Arduino) would make it easily integrable. Alongside this, developers can create specific C++ libraries or crates for each adapter type. For instance, a `SmartPlugAgent` crate would handle MQTT communication with the edge gateway and relay toggling, while an `IRShim` crate would manage pulse-width modulation (PWM) generation for IR LED drivers [58]. The edge gateway itself, also running Rust, would orchestrate these agents, aggregate their telemetry, and run the global safety logic for the household shard . The following table outlines the proposed firmware architecture, detailing the components, their roles, and the technologies involved.

Component	Role	Technologies / Protocols
Central Controller	Issues high-level `PdssIntents` to edge gateways.	City Fabric, Cloud Platform [189]
Edge Gateway (Per-Household)	Hosts the local `BugsLifeSafetyKernel`. Aggregates appliance adapters. Maintains the `DeterrentNodeShard`. Communicates with the city fabric via gated intents.	Rust, ALN, MQTT, CoAP [15] [135]
Smart-Plug Agent	Firmware on a smart plug. Receives intents, computes `safestep`, and toggles a solid-state relay. May include local power sensing.	Microcontroller (ESP32, etc.), Relay Driver, MQTT Client [58]
IR Shim Adapter	Firmware on a dedicated module. Learns IR command codes and re-transmits them based on `PdssIntents`. Acts as an IR repeater/emulator.	IR LED, Microcontroller with PWM [18] [106]
Panel-Relay Agent	Firmware on a module installed in a breaker panel. Controls power to a circuit or appliance branch. Provides strong isolation.	Optocoupled Relays, RS-485/Modbus for communication [98] [193]
Serial Bus Shim	Firmware on a module that taps into an appliance's internal bus (RS-485, I^2C). Interprets and injects control messages.	RS-485/I^2C Transceivers, Microcontroller [11] [192]
Adaptive Profile Module	Algorithm running on the Edge Gateway. Collects telemetry from external sensors. Fits conservative emission envelopes for new devices. Updates the appliance's shard.	Signal Processing (FFT for sound, etc.), Statistical Fitting [215]
LegacyAppliancePdssKernel-C++	Cross-platform C++ library implementing the core `PdssIntent` and `safestep` logic. To be used by all agent firmwares.	C++, Header-only design, No-std compatible [45]

This layered and modular architecture provides the flexibility and resilience needed to tackle the immense diversity of legacy hardware. By combining a universal safety kernel with a versatile suite of adapter patterns and intelligent learning mechanisms, the framework establishes a robust foundation for building a large-scale, adaptive, and equitable pest-deterrent infrastructure entirely from existing household assets.

Empirical Characterization and Data-Driven Governance

The theoretical elegance of an intent-based safety kernel and a universal adapter architecture is contingent upon empirical validation and a robust data governance framework. The proposed research direction mandates a rigorous, multi-stage process to move from abstract models to real-world deployment, ensuring that corridor definitions are grounded in physical reality and that the system's behavior remains predictable and safe. This process begins with systematic bench characterization of representative legacy appliances and extends through real-world "shadow" deployments and simulation, culminating in a formalized data structure for accountability and impact measurement.

The foundational research action is the **bench characterization of emission corridors** for major appliance classes . This involves selecting a statistically representative sample of devices—including box fans, mechanical-fridge freezers, window air conditioners, CRT/early LCD televisions, and analog audio systems—and subjecting them to controlled testing . Using programmable power supplies, data loggers, and calibrated sensors, researchers would map the relationship between an appliance's input (e.g., power supply voltage, control signal duty cycle) and its output emissions. Key measurements would include:

- **Acoustic Footprint:** Using a calibrated microphone array to capture the acoustic spectrum (dBA) at various duty cycles and operational modes [216]. This data helps define the `rnoise` coordinate and its associated corridor bands.
- **Surface Temperature Rise:** Employing infrared thermography to non-invasively measure the maximum temperature increase on exterior surfaces relative to ambient room temperature [126][147]. This provides the data needed to calculate the `rthermal` coordinate.
- **Structural Vibration:** Utilizing accelerometers to measure the root mean square (RMS) vibration levels transmitted through the floor or mounting surface [215]. This informs the `rvib` coordinate.
- **Energy Consumption:** Precisely measuring power draw to establish a baseline for calculating the system's energy overhead [132].

The dataset generated from these bench tests forms the empirical spine for establishing initial corridor bands (safe, gold, hard limits) for each appliance class . These bands are not arbitrary; they are calibrated against existing residential and occupational exposure standards, such as ASHRAE thermal comfort bands or ISO guidelines for mechanical vibration, ensuring that the system respects established norms for human health and safety [53] [54] [221]. For example, a `dT_hard` limit could be set to keep surfaces below 48°C (118°F), a threshold far below the point of material degradation or skin contact burn risk [130].

Once initial corridor models are established, the next phase is a real-world validation period conducted in a "silent" or "shadow" mode. In this stage, a pilot group of apartments would be instrumented with smart plugs, external sensors, and edge gateways . The firmware would communicate with the appliances and collect telemetry, but it would not yet be actuating them for pest deterrence. Instead, it would simulate hypothetical `PdssIntents`, logging the actual `rj` values produced by the appliances under real-world conditions, which may differ from the controlled lab environment due to factors like room acoustics, airflow obstructions, and appliance aging [177]. This

shadowing phase is critical for validating the bench-calibrated models and identifying discrepancies. Concurrently, researchers would develop and run simulations using JetsonLine-style modeling tools . These simulations would compute the trajectories of the Lyapunov residual, V_t, under various multi-appliance schedules, verifying that the global system remains stable and that the safety invariant $V_{t+1} \leq V_t$ holds true in a virtual environment before any real-world actuation occurs . Only after the shards demonstrate stable corridors and non-increasing residuals under a wide variety of simulated profiles would the project proceed to ethics-gated pilots involving actual actuation .

The culmination of this empirical process is the `qpudatashard`, a formal, persistent data structure that serves as the bedrock of the system's governance and accountability . The `LegacyAppliancePdssNodes2026v1.aln` file provides a canonical schema for recording every aspect of a node's lifecycle, from its initial characterization to its ongoing operation . This shard is more than just a data dump; it is a formal contract that enables several layers of governance. First, it facilitates automated decision-making through K/E/R (Knowledge/Eco-Risk) scoring gates. Before a device is deployed or a schedule is updated, the system checks if the relevant shard's K/E/R scores meet minimum thresholds (e.g., Knowledge-factor > 0.9, Eco-impact > 0.9, Risk-of-harm < 0.13) . This ensures that only adequately understood and environmentally beneficial configurations are permitted.

Second, the shard provides a transparent audit trail. Fields like `bostrom_tx_hex` and `evidencehex` anchor the node's configuration and safety-critical calculations (like its Lyapunov residual) to immutable, verifiable records on a distributed ledger . This creates a chain of custody for safety decisions, making the system's behavior auditable by regulators, facility managers, or residents themselves. Third, the shard enables city-level accounting and policy-making. Metrics such as `pdss_energy_kWh_per_day` allow for precise tracking of the system's operational cost, while the `baseline_pesticide_g_per_day_avoided` field quantifies the eco-positive impact . This metric can be calculated using CEIM-style mass-balance logic, adapted from water quality engineering, to estimate the total mass of active chemical ingredients displaced by the non-chemical deterrent overlay [81] [105]. This transforms an abstract sustainability goal into a tangible, creditable asset, akin to carbon offsets, which could be used to incentivize the adoption of the framework in municipal housing projects [34]. The table below summarizes the key governance and impact metrics tracked within the shard.

Metric Category	Shard Field(s)	Purpose and Rationale
Safety & Compliance	`rnoise`, `rthermal`, `rvib`, `residualVt`	Real-time monitoring of normalized risk coordinates and Lyapunov residual to enforce the `safestep` invariant.
Governance	`knowledgefactor01`, `ecoimpact01`, `riskofharm01`	Quantitative scores to gate deployment and updates, ensuring system maturity and positive environmental impact.
Operational Performance	`pdss_duty_fraction`, `pdss_energy_kWh_per_day`	Tracks the utilization and energy overhead of the PDSS system, ensuring it remains a light-load application.
Environmental Impact	`baseline_pesticide_g_per_day_avoided`	Estimates the mass of chemical pesticides displaced, providing a quantitative measure of eco-benefit.
Accountability & Audit	`bostrom_tx_hex`, `evidencehex`	Anchors safety-critical data to immutable, verifiable blockchain transactions, creating an auditable record of decisions.
Contextual Information	`applianceclass`, `adaptertype`, `ratedpowerW`	Metadata essential for correct interpretation of telemetry and application of appropriate corridor models.

By combining rigorous empirical characterization with a formal, data-driven governance model embodied in the `qpudatashard`, the research framework moves beyond a mere technical exercise. It establishes a socio-technical system that is not only safe and effective but also transparent, accountable, and capable of demonstrating its value in measurable terms. This approach builds trust and provides the necessary evidence base for scaling the technology from a niche research project to a widely adopted urban infrastructure solution.

City-Scale Integration and Strategic Implications for Equity and Sustainability

The successful deployment of a universal firmware-based pest-deterrent framework depends not only on its technical robustness but also on its strategic integration into the broader urban ecosystem and its commitment to social equity. The research plan explicitly prioritizes an "overlay" architecture, designed to integrate seamlessly with existing smart-city platforms and Building Management Systems (BMS) without necessitating a complete replacement of legacy infrastructure . This pragmatic approach acknowledges the heterogeneous nature of modern buildings, which often house a mix of IP-based and older bus-communication systems like BACnet, Modbus, or KNX [6] [10] [12]. The framework's ability to interoperate with these disparate systems is a key enabler of its scalability. Integration is achieved purely through the exchange of safety-gated intents and shard telemetry, meaning the city-side management platform interacts with the

framework at a high level of abstraction . The city dashboard would display aggregated risk metrics, energy consumption data, and estimated eco-benefits (like grams of pesticide avoided per building), but it would have no authority to override the local safety kernel residing on the edge gateway . This design ensures that the BugsLife system operates as a trusted, autonomous agent within the building, enhancing the capabilities of the existing BMS rather than competing with it. The Matter bridge specification provides a conceptual precedent for translating between proprietary data models and a unified application layer, a principle directly applicable to bridging the BugsLife intent model with various BAS protocols [5] [135].

A profound strategic implication of this framework is its potential to address long-standing inequities in urban infrastructure. Older, less expensive appliances are disproportionately located in lower-income buildings, which are also more likely to be part of public or social housing stock [121] [151]. There is a significant risk that such communities could be relegated to serving as "looser testbeds" for new technologies, receiving weaker safety oversight or fewer features due to the perceived unreliability of their underlying hardware. The research framework explicitly counters this risk through a "governance-by-design" approach. It proposes applying the same stringent K/E/R (Knowledge/Eco-Risk) gating rules and corridor strictness to "legacy-only" deployments as are applied to developments with modern, connected devices . In fact, the framework suggests that these retrofitted zones might warrant even stricter oversight to compensate for the lack of manufacturer documentation and cooperative support . The shard governance logic, which prevents deployment if $K<0.9$, $E<0.9$, or $R>0.13$, acts as an automatic safeguard, ensuring that no neighborhood is subjected to a less rigorously vetted version of the system simply because it has older appliances . This commitment to equal protection for all, regardless of the age of their hardware, is a cornerstone of the project's ethical framework and a critical factor for its long-term social license to operate.

Furthermore, the framework's emphasis on energy conservation and grid-aware scheduling aligns with broader goals of urban sustainability. While the direct energy consumption of modulating legacy appliances for deterrence is expected to be minimal—adding perhaps 18 Wh/day for a 60W fan running at 30% duty, a fraction of a typical household's appliance budget—it must still be considered in the context of the overall building energy load [132]. The framework promotes energy-conserving actuation by leveraging the natural thermal inertia of buildings and appliances [218]. For example, a window AC unit's compressor cycling can be scheduled during periods of lower electricity demand or when renewable energy generation is high, shifting energy use rather than necessarily reducing it. The `pdss_energy_kWh_per_day` metric in the shard provides the granular data needed for such analyses and for participating in transactive energy markets, where buildings can be compensated for providing demand flexibility to the grid

[52] [84]. This positions the collective fleet of legacy appliances not just as pest-deterrent nodes, but as a valuable, distributed energy resource.

Finally, the quantifiable eco-impact of the framework presents a powerful case for its adoption by municipalities and housing authorities. The ability to generate a karma-like ledger, tracking metrics like pesticide displacement at the building or block level, transforms an environmental benefit from a qualitative aspiration into a quantitative, auditable asset . Displacing even small amounts of potent rodenticides, measured in grams per day per building, accumulates to significant quantities at the city scale [105]. Across 1,000 buildings, displacing 0.5 g/day of pesticide results in approximately 183 kg/year of active chemical release avoided, a figure comparable to other high-impact EcoNet nodes . This data can be used to justify capital investment in retrofitting programs, attract funding from social impact investors focused on climate emergency activities, and inform public health and environmental policies [33] [34]. By re-purposing a massive, already-installed base of idle hardware, the framework embodies a circular economy approach, maximizing the utility of existing resources and minimizing electronic waste. The five buildable project ideas identified in the research plan—such as the `BugsLifeApplianceMapper` and `CityPdssOverlayPlanner`—provide the practical tools needed to translate this vision into action, enabling cities to identify optimal deployment zones and plan rollouts that maximize both ecological benefit and equitable access . In synthesis, the research framework extends beyond a novel technical solution; it represents a comprehensive socio-technical strategy for building safer, more sustainable, and more equitable urban futures by intelligently augmenting the world we already have.

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